

IMAT MOCK EXAM 3 — "Genetics & Molecular Rotation"

60 questions · 100 minutes · scoring: +1.5 correct, −0.4 wrong, 0 blank · personal target: 45/90.0 · Mock exam

Section A - Logical Reasoning & Reading (Q1-9)

1. "A gene is a region of DNA that encodes a functional product, usually a protein. In eukaryotes, coding regions (exons) are interrupted by non-coding regions (introns). After transcription, introns are removed and exons are spliced together to form mature messenger RNA, which is then translated by ribosomes."

Which statement is not supported?

- A)** Exons are coding regions.
- B)** Ribosomes transcribe DNA into RNA.
- C)** Introns are removed before translation.
- D)** A gene typically encodes a protein.
- E)** Splicing produces mature mRNA.

2. "A mutation is a heritable change in the nucleotide sequence of a genome. Mutations may be neutral, harmful, or — rarely — beneficial. Beneficial mutations can increase in frequency in a population over generations through natural selection, provided they confer a reproductive advantage."

The passage implies:

- A)** All mutations are harmful.
- B)** Neutral mutations are eliminated quickly.
- C)** Mutation rate determines fitness directly.
- D)** Natural selection acts on existing variation, including beneficial mutations.
- E)** Beneficial mutations always become fixed.

3. (General knowledge) Which Italian scientist won the 1986 Nobel Prize in Physiology or Medicine for the discovery of nerve growth factor (NGF)?

- A)** Camillo Golgi
- B)** Carlo Rubbia
- C)** Rita Levi-Montalcini
- D)** Salvador Luria
- E)** Renato Dulbecco

4. In a race, Ali finished before Bashir but after Chen. Dev finished last. Who finished first?

- A)** Chen
- B)** Ali
- C)** Dev
- D)** Cannot be determined
- E)** Bashir

5. "All enzymes are proteins." This statement is:

- A)** True only for digestive enzymes
- B)** Always true
- C)** True only in prokaryotes
- D)** Always false
- E)** Mostly true, but some RNA molecules (ribozymes) also catalyse reactions

6. In a certain language, "TREE" is written as "USFF". How is "BIRD" written?

- A)** CJTE **B)** AJSE **C)** CJSF **D)** BKSE **E)** CJSE

7. What comes next: 3, 6, 11, 18, 27, ?

- A)** 36 **B)** 38 **C)** 39 **D)** 40 **E)** 41

8. If today is Tuesday, what day will it be 100 days from today?

- A)** Monday **B)** Wednesday **C)** Tuesday **D)** Thursday **E)** Friday

9. A clock reads 7:00. What is the angle between the hour and minute hands?

- A)** 0° **B)** 90° **C)** 150° **D)** 180° **E)** 210°

Section B - Biology (Q10-32)

10. In DNA, the two strands run in opposite directions. This property is called:

- A)** Antiparallelism
B) Methylation
C) Supercoiling
D) Complementarity
E) Semiconservative

11. DNA replication is described as "semiconservative" because:

- A)** Both strands of the parent molecule are preserved together.
B) Replication is conservative in prokaryotes only.
C) Only the leading strand is copied.
D) Each new molecule consists of one parental strand and one newly synthesised strand.
E) The entire molecule is built anew.

12. The enzyme that joins Okazaki fragments by forming phosphodiester bonds is:

- A)** Helicase
B) Topoisomerase
C) DNA ligase
D) Primase
E) DNA polymerase III

13. During transcription in eukaryotes, the enzyme responsible is:

- A)** DNA polymerase
B) RNA polymerase
C) Ligase
D) Reverse transcriptase
E) Helicase

14. During which process are introns removed from pre-mRNA?

- A)** Splicing (post-transcriptional modification)
B) DNA repair
C) Translation
D) Transcription (initiation)
E) Replication

15. The three-nucleotide sequence on mRNA that specifies an amino acid is a:

- A)** Gene **B)** Anticodon **C)** Operon **D)** Promoter **E)** Codon

16. The anticodon is found on:

- A)** SnRNA **B)** mRNA **C)** DNA **D)** tRNA **E)** rRNA

17. The bonds linking amino acids in a protein are:

- A)** Peptide bonds
B) Hydrogen bonds
C) Phosphodiester bonds
D) Ester bonds
E) Glycosidic bonds

- 18.** The secondary structure of proteins (α -helix, β -sheet) is stabilised mainly by:
- A)** Peptide bonds
 - B)** Hydrogen bonds between backbone atoms
 - C)** Ionic bonds
 - D)** Hydrophobic interactions
 - E)** Disulfide bridges
- 19.** A diploid cell with $2n = 6$ undergoes meiosis. How many chromosomes are in each resulting gamete?
- A)** 2 **B)** 3 **C)** 4 **D)** 6 **E)** 12
- 20.** In a cross between two heterozygous tall plants ($Tt \times Tt$), where T (tall) is dominant, what is the expected phenotypic ratio in the offspring?
- A)** 9:3:3:1
 - B)** All tall
 - C)** 3:1 (tall : short)
 - D)** 1:2:1
 - E)** 1:1
- 21.** In humans, haemophilia is an X-linked recessive trait. A carrier mother ($X^H X^h$) and an unaffected father ($X^H Y$) have a son. What is the probability the son has haemophilia?
- A)** 0 **B)** 25% **C)** 50% **D)** 75% **E)** 100%
- 22.** A man who is colour-blind (X-linked recessive) has daughters with a woman who is not colour-blind and is not a carrier. What is true of their daughters?
- A)** None will carry the allele.
 - B)** All will be colour-blind.
 - C)** All will be colour-blind and carriers.
 - D)** 50% colour-blind, 50% carriers.
 - E)** All will be carriers (heterozygous) but not colour-blind.
- 23.** In a dihybrid cross ($AaBb \times AaBb$) with independent assortment, the expected phenotypic ratio is:
- A)** 1:1 **B)** 1:2:1 **C)** All dominant **D)** 3:1 **E)** 9:3:3:1
- 24.** A bacterium acquires antibiotic resistance via a plasmid from another bacterium. This is an example of:
- A)** Meiosis
 - B)** Mutation
 - C)** Mitosis
 - D)** Conjugation (horizontal gene transfer)
 - E)** Transcription
- 25.** The phases of mitosis in correct order are:
- A)** Prophase, metaphase, anaphase, telophase
 - B)** Prophase, anaphase, metaphase, telophase
 - C)** Telophase, anaphase, metaphase, prophase
 - D)** Metaphase, prophase, anaphase, telophase
 - E)** Anaphase, metaphase, prophase, telophase
- 26.** Which organelle contains digestive enzymes and functions in intracellular digestion?
- A)** Mitochondrion **B)** Chloroplast **C)** Nucleolus **D)** Ribosome **E)** Lysosome

27. Which structure is common to both plant and animal cells?
A) Plasmodesmata
B) Chloroplast
C) Mitochondrion
D) Large central vacuole
E) Cell wall
28. In cellular respiration, glycolysis produces a net gain of:
A) 0 ATP (all invested)
B) 2 ATP, 2 NADH, 2 pyruvate (per glucose)
C) 2 ATP, 2 FADH₂, 2 pyruvate
D) 4 ATP, 4 NADH, 2 pyruvate
E) 38 ATP
29. The non-protein component required for some enzymes to function (e.g., a metal ion) is called a:
A) Cofactor (or coenzyme if organic)
B) Apoenzyme
C) Competitive inhibitor
D) Substrate
E) Allosteric inhibitor
30. At the neuromuscular junction, the neurotransmitter released by motor neurons is:
A) GABA **B)** Serotonin **C)** Glutamate **D)** Acetylcholine **E)** Dopamine
31. The resting potential of a typical neuron is approximately:
A) -70 mV **B)** -30 mV **C)** 0 mV **D)** +30 mV **E)** +70 mV
32. Which vitamin is required for blood clotting?
A) Vitamin D **B)** Vitamin A **C)** Vitamin K **D)** Vitamin C **E)** Vitamin B12

Section C - Chemistry (Q33-47)

33. How many moles are in 36 g of water (H₂O; molar mass = 18 g/mol)?
A) 0.5 **B)** 1 **C)** 2 **D)** 18 **E)** 36
34. Balance: $__ \text{C}_3\text{H}_8 + __ \text{O}_2 \rightarrow __ \text{CO}_2 + __ \text{H}_2\text{O}$. The coefficient of O₂ is:
A) 3 **B)** 4 **C)** 5 **D)** 6 **E)** 8
35. The oxidation number of nitrogen in NO₃⁻ is:
A) -3 **B)** +3 **C)** +4 **D)** +5 **E)** +6
36. What is the pH of a 0.001 M solution of HNO₃ (strong acid)?
A) 1 **B)** 2 **C)** 3 **D)** 11 **E)** 13
37. Which of the following is a weak acid?
A) HNO₃
B) HBr
C) HCl
D) H₂SO₄
E) CH₃COOH (acetic acid)
38. Which bond is most polar?
A) C-H **B)** O-H **C)** H-H **D)** C-C **E)** N-H

39. Boyle's law states that, at constant temperature, the pressure of a fixed mass of gas is:
A) Independent of volume
B) Inversely proportional to its volume
C) Proportional to temperature
D) Directly proportional to its volume
E) Proportional to moles
40. The number of protons in an atom of ^{23}Na (sodium) is:
A) 11 **B)** 12 **C)** 22 **D)** 23 **E)** 34
41. Which of the following is an alkene?
A) C_3H_8 **B)** C_6H_6 **C)** CH_4 **D)** C_2H_4 **E)** C_2H_6
42. The functional group of alcohols is:
A) $-\text{CHO}$ **B)** $-\text{NH}_2$ **C)** $-\text{COOH}$ **D)** $-\text{CO}-$ **E)** $-\text{OH}$
43. Which has a $\text{C}=\text{O}$ group bonded to two carbons (within a chain)?
A) Carboxylic acid
B) Ketone
C) Alcohol
D) Aldehyde
E) Amine
44. Increasing the concentration of a reactant in a reaction generally:
A) Decreases activation energy
B) Decreases the rate
C) Shifts K_{eq}
D) Increases the rate (more frequent collisions)
E) Has no effect on rate
45. For an endothermic reaction, ΔH is:
A) Positive
B) Zero
C) Equal to activation energy
D) Negative
E) Always equal to ΔS
46. Which isotope is radioactive?
A) ^{13}C **B)** ^{16}O **C)** ^{14}N **D)** ^{12}C **E)** ^{14}C
47. A solution has a pH of 8. It is:
A) Strongly basic
B) Strongly acidic
C) Weakly acidic
D) Neutral
E) Weakly basic

Section D - Mathematics (Q48-54)

48. Simplify: $(2^3) \times (2^2)$.
A) $2^5 (= 32)$ **B)** 2^6 **C)** $2^1 (= 2)$ **D)** 4^5 **E)** 8
49. Solve: $\log_{10}(x) = 2$.
A) $x = 20$ **B)** $x = 100$ **C)** $x = 10$ **D)** $x = 2$ **E)** $x = 1000$
50. Solve: $x^2 = 49$ ($x > 0$).
A) 6 **B)** 7 **C)** 8 **D)** 14 **E)** 24.5

51. Solve the inequality: $x/3 - 1 \geq 2$.
A) $x \geq 6$ B) $x \geq 1$ C) $x \geq 12$ D) $x \geq 9$ E) $x \geq 3$
52. A square has perimeter 20 cm. What is its area?
A) 5 cm² B) 10 cm² C) 20 cm² D) 25 cm² E) 100 cm²
53. $\cos 60^\circ = ?$
A) 0 B) $\sqrt{3}/2$ C) $1/2$ D) $\sqrt{2}/2$ E) 1
54. Two fair coins are tossed. What is the probability of getting at least one head?
A) $1/4$ B) $1/2$ C) 1 D) $1/3$ E) $3/4$

Section E - Physics (Q55-60)

55. A car decelerates uniformly from 20 m/s to rest in 5 s. What is its deceleration?
A) 2 m/s² B) 4 m/s² C) 5 m/s² D) 10 m/s² E) 20 m/s²
56. Three 6 Ω resistors are connected in parallel. The equivalent resistance is:
A) 2 Ω B) 3 Ω C) 6 Ω D) 9 Ω E) 18 Ω
57. What is the voltage across a 10 Ω resistor carrying 0.5 A?
A) 0.05 V B) 5 V C) 10 V D) 20 V E) 50 V
58. A ball of mass 0.5 kg moves at 4 m/s. What is its kinetic energy?
A) 1 J B) 2 J C) 4 J D) 8 J E) 16 J
59. A 10 N force moves an object 5 m in the direction of the force. How much work is done?
A) 0 J B) 2 J C) 5 J D) 15 J E) 50 J
60. Specific heat capacity of water is 4200 J/(kg·°C). How much energy heats 2 kg of water by 10 °C?
A) 8,400 J B) 42,000 J C) 84,000 J D) 420,000 J E) 840 J

Answer key

This key refers to the option order printed in this paper. The online app shuffles option order on every attempt, so answer letters inside the app will differ from the ones here.

1	B	2	D	3	C	4	A	5	E
6	E	7	B	8	D	9	C	10	A
11	D	12	C	13	B	14	A	15	E
16	D	17	A	18	B	19	B	20	C
21	C	22	E	23	E	24	D	25	A
26	E	27	C	28	B	29	A	30	D
31	A	32	C	33	C	34	C	35	D
36	C	37	E	38	B	39	B	40	A
41	D	42	E	43	B	44	D	45	A
46	E	47	E	48	A	49	B	50	B
51	D	52	D	53	C	54	E	55	B
56	A	57	B	58	C	59	E	60	C